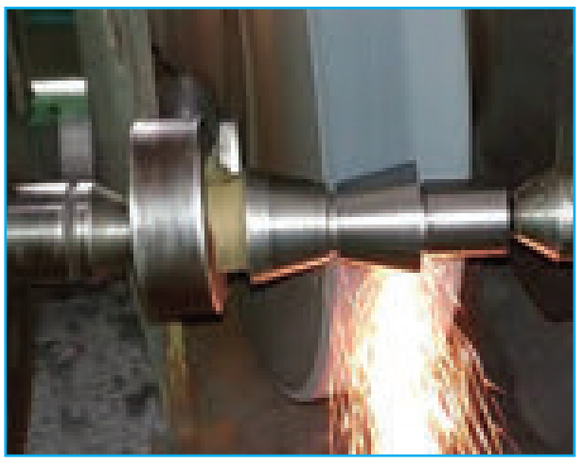
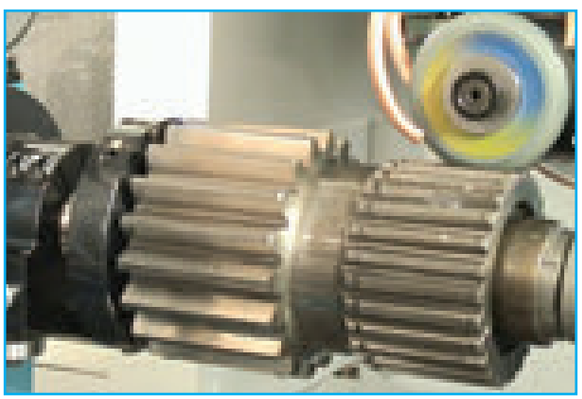


Form Grinding

Gear grinding

The teeth of gears are ground accurately on gear grinding machines for their shape. Gear grinding is done by the generating process or by using a form grinding wheel.

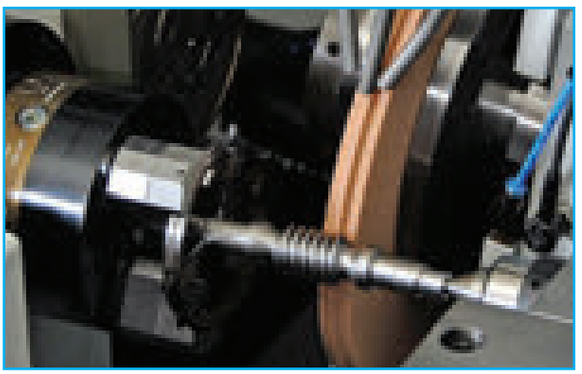


The generating process makes use of two saucer shaped grinding wheels. These wheels are used to grind two faces of successive teeth.

The forming process makes use of formed wheels to grind a tooth at a time. This is a very precise method of performing gear grinding.

Thread grinding

Thread grinding machines are used to grind threads accurately. The grinding wheel itself is shaped to the thread profile. These formed grinding wheels have one or multi threads on them.



Thread grinding

4.8 Wet and dry grinding

The method of spreading a good quantity of coolant over the work surface and wheel faces during grinding is known as 'wet grinding'. Soda water is used as a coolant. The process of grinding generates high amount of heat normally about 2000^0C. Various properties of the work material In order to reduce the heat generated during grinding, coolant is used. Wet grinding promotes long wheel life and better look of the ground surface. Coolant is pumped from the tank through pipelines.

Dry grinding is the method of doing grinding operation without applying coolant. Dry grinding produces undesirable effects on work surface. It leads to burring and discoloration of work surfaces. The cutting edges of the grinding wheel lose their cutting capacity. So, dry grinding should better be avoided.

4.9 Grinding wheel

A grinding wheel is a multi-tooth cutter made up thousands of hard particle known as abrasives having sharp edges. The abrasive grains are with a suitable bond, which act as a matrix to manufacture grinding wheels.

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